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ORIGINAL ARTICLE

Preliminary study on hand preference in captive northern white-cheeked gibbons (*Nomascus leucogenys*)

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Abstract The postural origin hypothesis and the task complexity hypothesis propose that hand preference in non-human primates evolved in association with body posture and task complexity, respectively. The results of previous studies testing these two hypotheses, however, vary greatly with the different primate species and methods used. To investigate the effect of body posture and task complexity on hand preference, we recorded bouts of hand usage in nine captive northern white-cheeked gibbons (*Nomascus leucogenys*) housed at Beijing Zoo as they reached for food items in a ground-reaching task, a box task, and a tube task. The results showed that four to seven of the nine gibbons displayed a hand preference at the individual level in different tasks, and that hand preference in individuals was task-specific; there was no group-level hand preference in any task. The box task seemed to elicit a greater strength of hand preference than the ground-reaching task at the individual level. Although the small sample size rules out drawing any strong conclusions concerning hand preference at the group level, our results suggest that the suspensory reaching posture might increase the expression of hand preference at the individual level. This study provides preliminary information on hand preference in captive northern white-cheeked gibbons, and will be helpful for future studies addressing the origin and evolution of hand preference in small apes.

Keywords Postural origin hypothesis · Task complexity hypothesis · Manual laterality · Suspensory reaching posture

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Introduction

Group-level manual laterality in humans is related to hemispheric functional lateralization and cognitive functions (Raymond and Pontier 2004; Chapelain and Hogervorst 2009). Hemispheric functional lateralization may improve cognitive abilities and behavioral efficiency by allowing simultaneous processing of different tasks, and by avoiding hemispheric competition through specific selective pressures (Corballis 1989; McGrew and Marchant 1997). The evolutionary origin and adaptive value of hemispheric functional lateralization in humans may be revealed by studying hand preferences in non-human primates (NHPs), since hand preference has been commonly used as an index of hemispheric functional specialization (Corballis 1989). A wealth of literature is available on hand preferences in primates, but the evolutionary origin of hand preference remains uncertain and the results are highly variable (Westergaard et al. 1997; Bennett et al. 2008; Hopkins et al. 2011; Morino 2011; Mangalam et al. 2015).

The postural origin hypothesis predicts that manual lateralization stems from an adaption to unimanual foraging in NHPs (MacNeilage 1987; Blois-Heulin et al. 2007) because body balance control and visual guidance (which manifests as increased cerebral lateralization) are required for individuals to transfer from a stable posture to an

unstable posture (Westergaard et al. 1998). For example, many arboreal primates display left-handedness for visually guided movements and manipulative actions; the right hand is used for postural control (MacNeilage 1987; Hopkins et al. 2011; Morino 2011; Zhao et al. 2012). In contrast, as their right hand is free to support the body, many terrestrial primates show right-handedness for precision and manipulative tasks (Hopkins et al. 2004, 2011; Bennett et al. 2008; Meguerditchian et al. 2013).

The task complexity hypothesis predicts that the direction and strength of hand preference at the group level are associated with the complexity of the task (Fagot and Vauclair 1991). A low-level task that requires less cognitive capability may lead to a symmetrical pattern of hand use, while a high-level task that demands fine motor acts, spatio-temporal and cognitive capabilities is more likely to elicit an asymmetrical pattern of hand use (Fagot and Vauclair 1991; Uomini 2009). High-level tasks, such as bimanual coordinated tube tasks, have been shown to elicit hand preference that is associated with neuroanatomical asymmetry within the primary motor cortex in chimpanzees (*Pan troglodytes*) (Hopkins and Cantalupo 2004). Increasing numbers of empirical studies have provided evidence to support the task complexity hypothesis (see Hopkins et al. 2011; Laurence et al. 2011; Zhao et al. 2012). Nonetheless, evidence opposing this hypothesis can also be found in studies on *Cercopithecus aethiops* by Harrison and Byrne (2000) and on *Macaca sylvanus* by Schmitt et al. (2008). Mangalam et al. (2015) recently proposed the niche structure hypothesis, stating that hand preference in NHPs might have evolved owing to structural and functional adaptations to feeding. These conflicting ideas about the origin of hand preference in NHPs may result from the use of different tasks, methods and study species by the researchers. A comprehensive test including both different body postures and different tasks in the same species might be helpful for revealing the origin of hand preference in NHPs. It is also desirable to widen the taxonomic coverage of hand preference studies in primate species.

The small apes are good models for investigating the evolution of hand preference in NHPs, due to their phylogenetic relationships with both Old World monkeys and great apes. Moreover, unlike great apes, small apes have a unique form of locomotion, brachiating or arm-swinging, as an adaptation to arboreal life (Carpenter 1976), and thus are good models in which to study the effects of body posture and task complexity on hand preference. Some studies have been conducted on hand preference in small apes (e.g., Olson et al. 1990; Redmond and Lamperez 2004; Morino 2011); however, these studies have used different tasks and different species, with variable results. For example, Olson et al. (1990) found that six captive white-handed gibbons showed no group-level hand preference in a simple task, but did show a group-level hand preference in a complex task. No group-level hand preference was found in brachiating simangs (*Symphalangus syndactylus*) (Redmond and Lamperez 2004) or northern white-cheeked gibbons (*Nomascus leucogenys*) (Barker 2008), but a group-level left-hand preference in wild siamangs engaged in a complex manual task was found (Morino 2011). Thus, considering both the effects of body posture and task complexity might help us to pinpoint the origin of hand preference in small apes.

The northern white-cheeked gibbon (*N. leucogenys*) is an endangered species of primate endemic to China, Laos and Vietnam. This gibbon was historically distributed in southwestern China, Northern Vietnam and Laos (Mootnick and Fan 2011). Data on the wild northern white-cheeked gibbon population in Vietnam and Laos are deficient (International Union for Conservation of Nature 2016). A recent study showed that this gibbon has become extinct in China due to illegal hunting, isolated small populations, and habitat fragmentation (Fan et al. 2013). There are about 30 individuals in Chinese zoos (Huang Bei, personal communication). In this study, we investigated white-cheeked gibbons' hand preference for taking food in three situations: a ground-reaching task, a box task and a tube task. The aims of this study were to test whether there is any hand preference at group level within each task, how body posture affects hand preference, and how task complexity affects hand preference.

Methods

Study subjects, management and husbandry

Our study subjects were two families of northern white-cheeked gibbons housed at Beijing Zoo, comprising nine individuals (four males and five females). The identification codes, ages in years and pedigree of the gibbons are shown in Fig. 1. The two families were housed separately in the Gibbons Exhibit and the Apes Exhibit enclosures at Beijing Zoo. Each enclosure consisted of an indoor cage (5 m in length, 4 m in width, and 3.5 m in height), and an outdoor cage (6 m in length, 5 m in width and 3.5 m in height), which was enriched with objects such as tree branches, iron chains, iron shelves, ropes, radiators, basins and cloth bags. The indoor and outdoor cages could be separated by a grille door. The zookeepers cleaned the enclosure at around 09:00 a.m. each day; this took about 0.5 h. Fruits, such as slices of apple, banana and orange, and grapes, were fed to the gibbons after the enclosure had been cleaned. Water was available ad libitum.

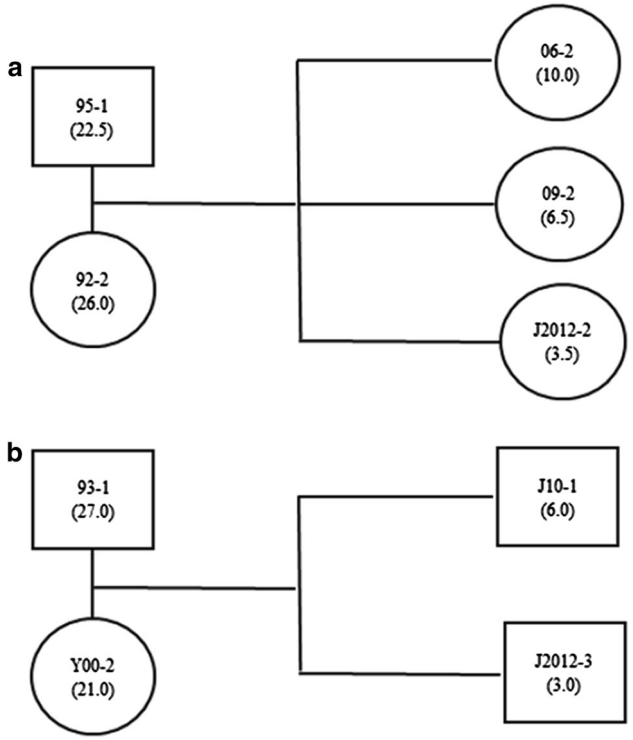


Fig. 1 Pedigree and age of the nine captive northern white-cheeked gibbons (*Nomascus leucogenys*) used in the study and kept in the Gibbons Exhibit (**a**) and the Apes Exhibit (**b**) at Beijing Zoo, People's Republic of China. Each gibbon's identification code is shown at the top inside the *square* (male) or *circle* (female); the bottom number *in parentheses* is the gibbon's age in years at the time of the study

According to the breeding records kept by the zoo, female gibbons usually first give birth at around 7 years of age, and males reach sexual maturity at around 6 years of age. Thus, we defined the subjects over 7 years old as adults, and younger subjects as juveniles/sub-adults. The mean age (±SE) of the gibbons was 13.94 ± 3.34 years (range 3–27 years, $n = 9$).

Testing procedures

We investigated the nine gibbons' hand preferences when they were reaching for food in three separate tasks: the ground-reaching task, the box task and the tube task, following the manual preference paradigm of Fagot and Vauclair (1991). In brief, we repeatedly presented a given task to the same individuals, observed their hand preference when reaching for food, and then calculated each individual's value for right- and left-hand uses. To run each task test, we placed food items in different places or inside containers, creating tasks of different levels of complexity. This approach to measuring hand preference has been used with several species of NHPs [*Pan paniscus* (Hopkins 1993); *Cebus paella* (Westergaard et al. 1997); *Macaca mulatta* (Westergaard et al. 1998); *Lophocebus albigena* (Blois-Heulin et al. 2007)]. The ground-reaching task is a spontaneous and simple, routinely unimanual feeding action with a stable (sitting) body posture. The box task is a unimanual reaching task with an unstable (suspensory) body posture (Pan et al. 2011). The tube task is a standard bimanual coordination task that has been used to evaluate hand preference in many NHPs (Hopkins 1995; Nelson et al. 2015).

To conduct the ground-reaching-task test, the zookeeper placed food items (slices of apple, banana and orange, grapes, etc.) on the ground in the indoor cage, at the normal feeding time in the morning. Then, we directly observed which hand the gibbons used when they reached for food items on the ground (Blois-Heulin et al. 2007) and recorded them with a digital video camera (Sony HDR-XR260; Sony, Shanghai). Each individual was observed for 20 min on each day of data collection. Data for the ground-reaching task were collected from 11 March to 10 May 2014, from 2 to 25 January 2015, and 18–30 March 2016.

For the box task test, we used a wooden box (50 cm × 25 cm × 25 cm) with a hole at the front (diameter 10 cm; Fig. 2a). The box had been used as an environmental enrichment item for the gibbons, and we trained them to take food items from the box during 3 consecutive days prior to data collection. The box was

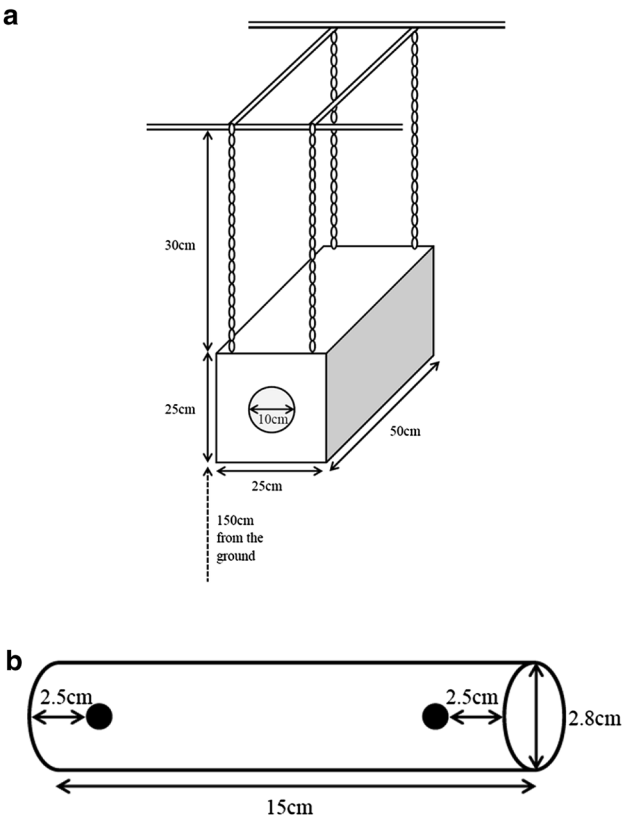


Fig. 2 Diagram of the box (**a**) and tube (**b**) used in the hand-preference test for gibbons

hung with iron chains 30 cm under the indoor iron shelves and 150 cm above the ground (Fig. 2a), to elicit reaching behaviors with a suspensory posture. To conduct the test, the zookeeper placed food items in the wooden box at the normal feeding time in the morning. We started observation and video recording when the gibbons began to reach into the box to remove the food items. Usually, the gibbons used one hand to maintain their body posture, and the other hand to reach for the food inside the box. We used the same observational and recording method and digital video camera used in the ground-reaching task. We conducted the box test during the periods between 11 March and 10 May 2014, and between 2 and 25 January 2015. Each observation lasted for 20 min.

The tube task, first used by Hopkins (1995), is commonly used to evaluate hand preference in NHPs (Nelson et al. 2015). For this test we used peanut butter to stick slices of banana to the inner surface of polyvinylchloride (PVC) tubes (2.8 cm in internal diameter and 15 cm in length). Two slices were placed in each tube, one at ~2.5 cm from each end of the tube (Fig. 2b). We trained the gibbons to take food items from the tubes during 5 habituation days prior to data collection. During the tube task, the gibbons held the PVC tube with one hand and retrieved food from the tube with the other hand. For each test, at least six tubes were placed on the ground of the indoor cage simultaneously so as to minimize competition for access. The observational and recording methods and video camera were the same as those used in the ground-reaching and box tasks. We conducted the tube task from 26 February to 31 March 2016. Each observation lasted 20 min.

To reduce the possible effects of food item size and shape on hand-preference expression, zookeepers cut the larger food items into small slices of similar size, and used whole grapes. All animals were healthy during the study period. Our protocol had no adverse effects on gibbons, and was approved by the Committee of Animal Welfare and Ethic of Beijing Normal University (no. CLS-EAW-2014-014) and Beijing Zoo.

Data collection

We collected data on each gibbon's hand preference using focal animal sampling and continuous recording methods (Martin and Bateson 1993). Using the video recordings for each task, bouts of hand usage by each gibbon were coded by a person (C. Y. Liu) who was blind to the experimental design and each gibbon's identity. We defined a bout of hand use as the first reaching or extracting action (Hopkins et al. 2001). A new bout was recorded when the focal animal moved to another place inside the cage, picked up a new tube, changed the orientation of a tube, or changed hands when taking food from the tube or box (Zhao et al. 2012). For each of the three tasks, we defined the hand used to reach for and take the food items as the dominant hand. To keep consistency of body posture, we collected data for the ground-reaching task and tube task only when the gibbons took food while sitting on the ground or on a tree branch. If the focal subject was disturbed, for example by another gibbon stealing a food item from the focal subject, the observation was excluded from the analysis.

Data analysis

To quantify hand preference at the individual level, we first used binomial tests to evaluate whether there were significant differences between left- and right-hand uses for each individual within each task (Schweitzer et al. 2007). Next, we calculated the handedness index (HI) to indicate the hand preference for each individual in each task test, to quantify individual laterality bias. The HI was calculated by the formula: $HI = (R-L)/(R + L)$, where L is the number of bouts in which the left hand was used, and R is the number of bouts in which the right hand was used (Hopkins 1999). The HI ranges from -1 (completely left-handed) to $+1$ (completely right-handed). We used absolute values of HI (ABS-HI) to quantify the strength of individual hand preference regardless of the direction of the hand preference (Schweitzer et al. 2007).

We used Spearman correlation tests to evaluate the relationships between the bouts of hand usage and the values for HI or ABS-HI, in order to test the reliability of our data prior to further statistics within each task (Zhao et al. 2012). To examine the stability of hand preference for each task, we divided all data into two equal time periods, i.e., an earlier half and a later half, regardless of the year of observation. Then we used Spearman correlation tests to evaluate hand-preference stability between the two groups within each task.

We used χ^2-tests to evaluate whether left- or right-hand preference existed at the group level in the ground-reaching, box and tube tasks. To test the postural origin hypothesis, we used Wilcoxon signed rank tests to test for differences in HI and ABS-HI between the ground-reaching task and the box task. Finally, we used Wilcoxon signed rank tests to test for task-complexity differences in the direction/strength of hand preference between the ground-reaching and tube tasks. We used IBM SPSS 20.0 (SPSS, Chicago, IL) to analyze all data. All tests were two-tailed, and the significance level was 0.05.

Results

All nine gibbons completed the ground-reaching task, box task and tube task. In total, we collected 71 video episodes for each of the ground-reaching task and the box task, and

48 for the tube task. The mean number of hand-usage bouts per subject was 387 ± 119 (range 106–915, $n = 9$) for the ground-reaching task, 137 ± 18 (range 55–220, $n = 9$) for the box task, and 190 ± 31 (range 53–333, $n = 9$) for the tube task. The number of bouts per subject was not significantly correlated with the HI (ground-reaching task $r = -0.033$, $n = 9$, $P = 0.932$; box task $r = 0.083$, $n = 9$, $P = 0.831$; tube task $r = -0.050$, $n = 9$, $P = 0.898$) or the ABS-HI (ground-reaching task $r = 0.100$, $n = 9$, $P = 0.798$; box task $r = 0.300$, $n = 9$, $P = 0.433$; tube task $r = -0.117$, $n = 9$, $P = 0.765$). These results suggest that the varying numbers of bouts of hand usage did not skew the distribution of the HI.

Combining adult and juvenile data, there was no significant correlation between HI in the two time periods for any task (ground-reaching task $r = 0.183$, $n = 9$, $P = 0.637$; box task $r = 0.192$, $n = 9$, $P = 0.620$; tube task $r = 0.667$, $n = 9$, $P = 0.050$). For just the five adults, however, we found significant positive correlations between HI in the two time periods only for the box task ($r = 0.900$, $P = 0.037$) and the tube task ($r = 0.900$, $P = 0.037$), but not for the ground-reaching task ($r = 0.400$, $P = 0.505$). For the four juveniles, we found no correlation between the HI in the two time periods for any task (ground-reaching task $r = 0.400$, $P = 0.600$; box task $r = -0.200$, $P = 0.800$; tube task $r = 0.200$, $P = 0.800$). These results suggest that the expression of hand preference was unstable in juveniles for all tasks, but stable in adults for either box task or tube task.

We found one right-handed and three left-handed gibbons in the ground-reaching task test, three left-handed and three right-handed gibbons in the box task test, and three left-handed and four right-handed gibbons in the tube task (Table 1). Five, three and two gibbons were classed as ambidextrous for the ground-reaching task, box task and tube task, respectively. Five gibbons changed their dominant hand either from right to left or vice versa from the box task to the tube task, and one changed from ambidextrous to left-handed. However, there was no hand preference at the group level in any task (ground-reaching task $\chi^2 = 0.667$, $df = 2$, $P = 0.717$; box task $\chi^2 = 0$, $df = 2$, $P > 0.999$; tube task $\chi^2 = 0.667$, $df = 2$, $P = 0.717$).

We found no significant difference in HIs between the ground-reaching task and the box task ($z = 0.114$, $df = 8$, $P = 0.953$), but the strength of hand preference (ABS-HI) in the box task was significantly greater than in the ground-reaching task ($z = -2.666$, $df = 8$, $P = 0.008$). There were no significant differences in either direction ($z = -0.178$, $df = 8$, $P = 0.859$) or strength ($z = -1.599$, $df = 8$, $P = 0.110$) of hand preference between the ground-reaching task and the tube task.

Table 1 Data on hand preference for the ground-reaching task, box task and tube task in nine captive northern white-cheeked gibbons

Subjects	Sex	Age class	Ground-reaching task						Box task						Tube task					
			L	*R*	Total	*P*	HI[a]	Category[b]	*L*	*R*	Total	*P*	HI[a]	Category[b]	*L*	*R*	Total	*P*	HI[a]	Category[b]
93-1	Male	Adult	285	548	733	0.000	0.316	Right	45	175	220	0.000	0.591	Right	208	0	208	0.000	−1	Left
95-1	Male	Adult	130	39	169	0.000	−0.538	Left	88	16	104	0.000	−0.692	Left	97	200	297	0.000	0.347	Right
92-2	Female	Adult	65	74	139	0.498	0.065	Ambidextrous	132	21	153	0.000	−0.725	Left	4	114	118	0.000	0.932	Right
Y00-2	Female	Adult	524	391	915	0.000	−0.145	Left	59	83	152	0.053	0.169	Ambidextrous	200	133	333	0.000	−0.201	Left
06-2	Female	Adult	64	62	126	0.929	−0.016	Ambidextrous	51	64	115	0.263	0.113	Ambidextrous	86	65	151	0.103	−0.139	Ambidextrous
j10-1	Male	Juvenile	432	395	827	0.211	−0.045	Ambidextrous	22	66	88	0.000	0.5	Right	42	11	53	0.000	−0.585	Left
j2012-3	Male	Juvenile	124	123	247	1.000	−0.004	Ambidextrous	35	20	55	0.058	−0.273	Ambidextrous	65	67	132	0.931	0.015	Ambidextrous
09-2	Female	Juvenile	77	48	125	0.012	−0.232	Left	56	96	152	0.001	0.263	Right	117	151	268	0.044	0.127	Right
j2012-2	Female	Juvenile	49	57	106	0.497	0.075	Ambidextrous	164	36	200	0.000	−0.64	Left	60	89	149	0.021	0.195	Right
Mean (SE)					387 (119)		−0.058 (0.079)				137 (18)		−0.077 (0.173)				190 (31)		−0.034 (0.184)	

L Mean bouts of usage of the left hand, *R* mean bouts of usage of the right hand, HI handedness index

$P \geq 0.05$ indicates ambidextrous, $P < 0.05$ indicates left- (or right)-handed

[a] Calculated via a commonly used formula: $(R^b - L^a)/(R^b + L^a)$

[d] Based on the *P*-value of the binomial test and HI

Discussion

This is, to our knowledge, the first study on the effects of both body posture and task complexity on hand preference in the *Nomascus* genus. We found that the number of observed bouts of hand usage was unrelated to the distribution of hand preference in the three tasks. This result confirmed the reliability of our measurements of hand preference, and is highly consistent with those previously reported by other researchers (Zhao et al. 2012; Hopkins 2013). Hand-preference expression was stable in adults only for the box task and tube task, and not for the ground-reaching task. However, hand-preference expression in juveniles was not stable for any of the three tasks. This suggests that the stability of hand-preference expression of these captive northern white-cheeked gibbons might be age and task specific. Morino (2011) reported that hand preferences of juveniles were significantly more variable than those of adults in wild siamangs. Schaafsma et al. (2009) suggested that an increase in hand preference with age in humans was related to either social pressure or adaptation. There was no group-level hand preference in any tasks in the northern white-cheeked gibbons, yet we found that the number of individuals showing hand preference increased with task complexity, from four in the ground-reaching task, to six in the box task and seven in the tube task. Moreover, the hand preferences of individuals were not consistent across different tasks.

Previous studies in small apes have shown that hand preference is closely related to body posture and task complexity. Siamangs showed no group-level hand preference during brachiation (Redmond and Lamperez 2004), and a similar result was found in white-cheeked gibbons (Barker 2008). However, Olson et al. (1990) found that six captive white-handed gibbons consistently used their left hands to reach for raisins that were affixed to a mesh fence, even though they did not show group-level hand preference when they picked raisins up from the ground. A group-level left-hand preference was also reported in wild siamangs engaged in a complex manual task (Morino 2011). Our finding of no group-level hand preference in northern white-cheeked gibbons is consistent with results of Redmond and Lamperez (2004) and Barker (2008), but not with those of Olson et al. (1990) and Morino (2011). The differences might be related to the small sample size in our study (Hopkins 2006). Due to the limited availability of our study species in captivity in China, we could not test the minimum sample size of ten proposed by Hopkins (2006) for hand-preference studies in primates. Moreover, other small-scale studies on hand preference have revealed stronger hand preferences, albeit without showing group-level preference (Lilak and Phillips 2008; Chapelain and Hogervorst 2009). These results indicate that hand-preference expression in NHPs might be related to other factors, such as structural and functional adaptations to feeding (Mangalam et al. 2015), in addition to species differences and issues relating to sample size.

The nine gibbons displayed a stronger hand preference in the box task than in the ground-reaching task, and there was no significant difference in the direction of hand preference. These results are in agreement with the postural origin hypothesis, which proposes that an unstable posture may lead to stronger hand preference because greater balance control and more visual guidance are required to transfer from a stable posture (e.g., sitting) to an unstable posture (e.g., suspensory) (Westergaard et al. 1998). Small-bodied gibbons are typical arboreal species, thus hand-preference strength was predicted to increase when a posture was adopted requiring the animal to maintain body balance (De Vleeschouwer et al. 1995). The box task in our study might stimulate the gibbons to reach for food from a suspensory posture, thus requiring more motor precision and more visual guidance (Fagot and Vauclair 1991). Previous studies have shown that unstable postures result in increased preferred hand use in NHPs including white-handed gibbons (Olson et al. 1990), tufted capuchins (*Cebus apella*) (Westergaard et al. 1997), Campbell's monkeys (*Cercopithecus campbelli*) (Chapelain et al. 2006), and chimpanzees (Braccini et al. 2010). Our data suggest that body posture influences hand-preference strength at the individual level in the northern white-cheeked gibbon.

A previous study showed that a group-level hand preference is more likely to emerge when small apes perform complex tasks (Olson et al. 1990). Given its requirement for fine motor acts and cognitive capacities the bimanual tube task is more likely than simple spontaneous feeding to induce hand preferences (Colell et al. 1995; Blois-Heulin et al. 2006). The tube task has induced significant group-level hand preference in some NHP species [*Pan troglodytes* (Hopkins et al. 2004, 2011); *Pongo pygmaeus* (Hopkins et al. 2004, 2011); *Cercopithecus neglectus* (Schweitzer et al. 2007); *Gorilla gorilla* (Meguerditchian et al. 2010); *Pan paniscus* (Hopkins et al. 2011); *Cercocebus torquatus torquatus* (Laurence et al. 2011), and *Rhinopithecus roxellana* (Zhao et al. 2012)]. The ground-reaching task is a spontaneous feeding action which requires less cognitive capacity, and so it may lead to symmetrical patterns of hand use in NHPs (Fagot and Vauclair 1991; Uomini 2009). We found that more individual gibbons showed laterality biases (either right- or left-handedness) in the tube task than in the ground-reaching task, but there was no group-level hand preference (see Table 1). This result suggests that these small apes might be either right-handed or left-handed (or indeed

ambidextrous) when they carry out coordinated bimanual tasks. In terms of direction or strength of hand preference, our data failed to show significant differences between the ground-reaching task and the tube task. A similar result was previously reported in the Barbary macaque (*Macaca sylvanus*) (Schmitt et al. 2008). Moreover, chimpanzees showed no group-level hand preference in the highly complex task of fishing for ants (Marchant and McGrew 2007). These results indicate that expression of group-level hand preference may be more variable across different species and tasks than initially assumed (Schmitt et al. 2008).

Six of the nine gibbons showed an opposite directional hand preference in the tube and box tasks. In the latter, the gibbons used one hand (the hand providing physical strength) to keep their body stable, while using the other hand for skillful manipulation reaching for food from the box. However, when performing the tube task on the ground or in branches from a sitting posture, the hand used to provide physical strength was free, and was used for manipulation. These observations may indicate that different divisions of labor in hand usage exist for each individual in three-dimensional space, and in situations in which they are required to keep their body stable (Mangalam et al. 2014).

In summary, our study adds information on the northern white-cheeked gibbon *N. leucogenys* to the literature concerning manual lateralization. We found that four to seven of our nine gibbons displayed individual-level hand preferences within each task, but there was no group-level hand preference in any tasks. Moreover, we found a stronger hand preference in the box task than in the ground-reaching task, with no difference in hand-preference direction. Our data do not support the task complexity hypothesis, and the gibbons showed task-specific hand preference. A larger sample size is needed to more fully address the origin of hand preference in this endangered species of gibbon.

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Compliance with ethical standards

All procedures performed in studies involving animals were in accordance with the ethical standards of the institution or practice at which the studies were conducted. All applicable international, national, and/or institutional guidelines for the care and use of animals were followed. This article does not contain any studies with human participants performed by any of the authors.

Conflict of interest None.

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